

Tabla VLGM

Red

1ª ipasis

192.23.55.0

192.23.55.1

192.23.55.62

192.23.55.64

192.23.55.65

192.23.55.94

192.23.55.95

C3

192.23.55.96

192.23.55.97

192.23.55.110

192.23.55.111

Carret: 00013324

A: 01-223

B: 33-55

Red base asignada: 10.23.55.0/24

captura de:

redes

interfases

ip disposicion

7 trase de ip

2 ejemplar

1. 10.23.55.0/24

1.1 Redes ordenadas

Red Hosts

Pc1 50

Pc2 25

Pc3 12

Pc4 6

R1-R2 2

R2-R3 2

1.2 Datos de cada red

① Pc1: $2^n \rightarrow 2^6 = 64$
 $64 - 2 = 62 \text{ hosts}$

Máscara: $32 - 6 = /26$

Salto: $256 - 192 = 64$

③ Pc3: $2^n \rightarrow 2^4 = 16$
 $16 - 2 = 14 \text{ hosts}$

Máscara: $32 - 4 = /28$

Salto: $256 - 240 = 16$

② Pc2: $2^n \rightarrow 2^5 = 32$
 $32 - 2 = 30 \text{ hosts}$

Máscara: $32 - 5 = /27$

Salto: $256 - 224 = 32$

④ Pc4: $2^n \rightarrow 2^3 = 8$
 $8 - 2 = 6 \text{ hosts}$

Máscara: $32 - 3 = /29$

Salto: $256 - 248 = 8$

Tabla VLGM

Red

1ª ip asign

192.23.55.0

192.23.55.1

192.23.55.62

192.23.55.64

192.23.55.65

192.23.55.94

192.23.55.96

192.23.55.97

192.23.55.110

Pc4

192.23.55.112

192.23.55.113

192.23.55.116

⑤ R1-R2

$$2^n - 2 = 4$$

$$4 - 2 = 2 \text{ hosts}$$

$$\text{Máscara: } 32 - 2 = 30$$

$$\text{Salto: } 256 - 252 = 4$$

⑥ R2-R3

$$2^n - 2 = 4$$

$$4 - 2 = 2 \text{ hosts}$$

$$\text{Máscara: } 32 - 2 = 30$$

$$\text{Salto: } 256 - 252 = 4$$

1.3 Tabla VLGM

Red

1ª ip asignada

Última ip asignada

Broadcast

Pc1

10.23.55.0

10.23.55.1

10.23.55.62

10.23.55.63

Pc2

10.23.55.64

10.23.55.65

10.23.55.94

10.23.55.95

Pc3

10.23.55.96

10.23.55.97

10.23.55.110

10.23.55.111

Pc4

10.23.55.112

10.23.55.113

10.23.55.118

10.23.55.119

R1-R2

10.23.55.120

10.23.55.121

10.23.55.122

10.23.55.123

R2-R3

10.23.55.124

10.23.55.125

10.23.55.126

10.23.55.127

1.4 Asignación de puertos

Conexión

Puertos

Ip asignada

Gateway

R1

Fa 0/0

10.23.55.1

—

Pc1

E/0

10.23.55.2

10.23.55.1

R1

Fa 0/1

10.23.55.65

—

Pc2

E/0

10.23.55.66

10.23.55.65

R1

Fa 1/0

10.23.55.97

—

Pc3

E/0

10.23.55.98

10.23.55.97

R3

Fa 0/0

10.23.55.113

—

Pc4

E/0

10.23.55.114

10.23.55.113

R1	Fa 2/0	10.23.55.121	—
R2	Fa 0/0	10.23.55.122	—
R2	Fa 0/1	10.23.55.125	—
R3	Fa 0/1	10.23.55.126	—

-Ejercicio 2-

Carrete: 000 13324

A: 01 → 23

B: 33 → 55

Red base asignada: 192.23.55.0/24

2.1 Redes ordenadas

Red	Hosts		
Pc1	50	R3-R4	2
Pc2	25	R4-R5	2
Pc3	12	R5-R2	2
Pc4	8		
Pc5	6		
R1-R3	2		
R1-R2	2		
R3-R2	2		

2.2 Datos de cada red

- ① P_{c1} : $2^n \rightarrow 2^6 = 64$
 $64 - 2 = 62 \text{ hosts}$
Máscara: $32 - 6 = /26$
Salto: $256 - 192 = 64$
- ② P_{c2} : $2^n \rightarrow 2^5 = 32$
 $32 - 2 = 30 \text{ hosts}$
Máscara: $32 - 5 = /27$
Salto: $256 - 224 = 32$
- ③ P_{c3} : $2^n \rightarrow 2^4 = 16$
 $16 - 2 = 14 \text{ hosts}$
Máscara: $32 - 4 = /28$
Salto: $256 - 240 = 16$
- ④ P_{c4} : $2^n \rightarrow 2^4 = 16$
 $16 - 2 = 14 \text{ hosts}$
Máscara: $32 - 4 = /28$
Salto: $256 - 240 = 16$
- ⑤ P_{c5} : $2^n \rightarrow 2^3 = 8$
 $8 - 2 = 6 \text{ hosts}$
Máscara: $32 - 3 = /29$
Salto: $256 - 248 = 8$
- ⑥ $R_1 - R_3$: $2^n \rightarrow 2^2 = 4$
 $4 - 2 = 2 \text{ hosts}$
Máscara: $32 - 2 = /30$
Salto: $256 - 252 = 4$
- ⑦ $R_1 - R_2$: $2^n \rightarrow 2^2 = 4$
 $4 - 2 = 2 \text{ hosts}$
Máscara: $32 - 2 = /30$
Salto: $256 - 252 = 4$
- ⑧ $R_3 - R_2$: $2^n \rightarrow 2^2 = 4$
 $4 - 2 = 2 \text{ hosts}$
Máscara: $32 - 2 = /30$
Salto: $256 - 252 = 4$
- ⑨ $R_3 - R_4$: $2^n \rightarrow 2^2 = 4$
 $4 - 2 = 2 \text{ hosts}$
Máscara: $32 - 2 = /30$
Salto: $256 - 252 = 4$
- ⑩ $R_4 - R_5$: $2^n \rightarrow 2^2 = 4$
 $4 - 2 = 2 \text{ hosts}$
Máscara: $32 - 2 = /30$
Salto: $256 - 252 = 4$
- ⑪ $R_5 - R_2$: $2^n \rightarrow 2^2 = 4$
 $4 - 2 = 2 \text{ hosts}$
Máscara: $32 - 2 = /30$
Salto: $256 - 252 = 4$

2.3 Wildcard

2.3 Tabla VLGM

Red	1 ^{ra} ip asignada	Última ip asignada	Broadcast
Pc1	192.23.55.0	192.23.55.1	192.23.55.63
Pc2	192.23.55.64	192.23.55.65	192.23.55.95
Pc3	192.23.55.96	192.23.55.97	192.23.55.119
Pc4	192.23.55.112	192.23.55.113	192.23.55.122
Pc6	192.23.55.128	192.23.55.129	192.23.55.135
R1-R3	192.23.55.136	192.23.55.137	192.23.55.139
R1-R2	192.23.55.140	192.23.55.141	192.23.55.143
R3-R2	192.23.55.144	192.23.55.145	192.23.55.147
R3-R4	192.23.55.148	192.23.55.149	192.23.55.151
R4-R5	192.23.55.152	192.23.55.153	192.23.55.155
R5-R2	192.23.55.156	192.23.55.157	192.23.55.159

4 Asignación de puertos

Conexión	Puertos	Ip asignada	Gateway
R5	Fa0/0	192.23.55.1	—
Pc1	E/0	192.23.55.2	192.23.55.1
R2	Fa0/0	192.23.55.65	—
Pc2	E/0	192.23.55.66	192.23.55.65

Q3	Fa 0/0	192.23.55.92	—
Pc3	E/0	192.23.55.98	192.23.55.97
R1	Fa 0/0	192.23.55.113	—
Pc4	E/0	192.23.55.114	192.23.55.113
R4	Fa 0/0	192.23.55.120	—
Pc6	E/0	192.23.55.130	192.23.55.129
R1	Fa 0/1	192.23.55.137	—
R3	Fa 0/1	192.23.55.138	—
R1	Fa 1/0	192.23.55.141	—
Q2	Fa 0/1	192.23.55.142	—
P-3	Fa 1/0	192.23.55.145	—
R2	Fa 1/0	192.23.55.146	—
R3	Fa 2/0	192.23.55.149	—
Q4	Fa 0/1	192.23.55.150	—
R4	Fa 1/0	192.23.55.153	—
R5	Fa 0/1	192.23.55.154	—
Q5	Fa 1/0	192.23.55.157	—
R2	Fa 2/0	192.23.55.158	—

2.5 Wildcard masks :

$$P_{C4} : 255 - 240 = 15$$

$$P_{C7} - P_{C3} : 255 - 252 = 3$$

↳ Aplica para toda conexión de canto

$$P_{C2} : 255 - 224 = 31$$

$$P_{C3} : 255 - 240 = 15$$

$$P_{C6} : 255 - 248 = 7$$

$$P_{C7} : 255 - 192 = 63$$